# Painting with Darts: Fine-Tuning $\pi_{0.7}$ for Language-Conditioned Aiming on an SO-101

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**Summary.** We fine-tune $\pi_{0.7}$ [1] to aim a motorized dart blaster held by an SO-101 arm at any cell of a 5×5 grid, given the instruction `shoot` ⟨`cell`⟩. A language model (Claude) sits above the policy, as in hierarchical VLA systems [2]: it chooses cells to draw a picture, checks the board after each shot and re-issues misses. The policy is trained on 116 teleoperated demonstrations (16 minutes) collected in one session. We release the data in LeRobot format [3], with the code, wiring and all fine-tuning logs.

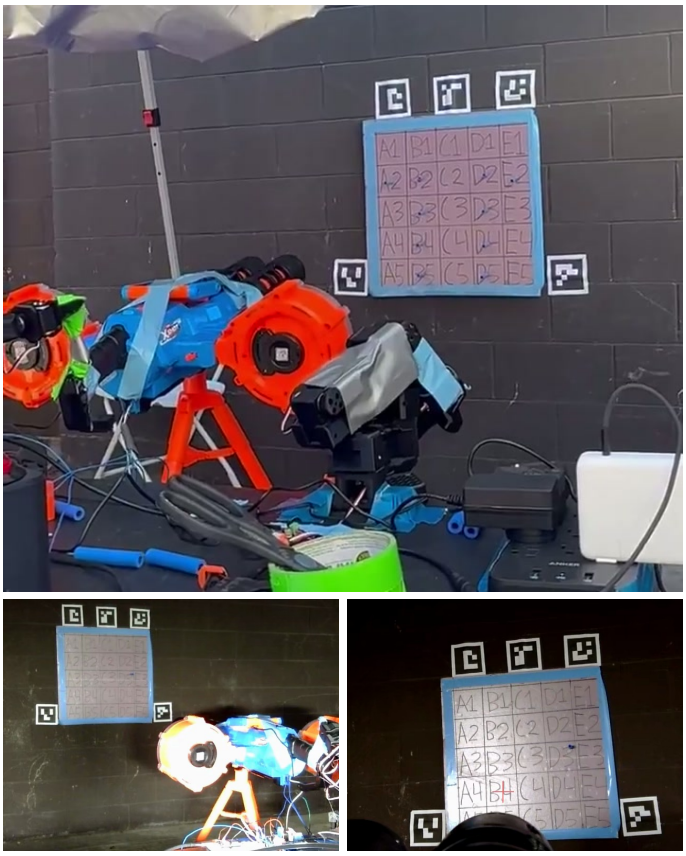


Fig. 1. Top: the rig. Bottom: the two policy views during a `shoot B4` demonstration, rear camera (left) and blaster-mounted camera with the barrel cross (right).

## Setup

The gripper is replaced by a wire loop around the blaster handle, and an ESP32-driven relay replaces the trigger switch, so firing is a binary action. Observations are two 640×480 images and five joint angles at 30 Hz; actions are five leader-arm joint targets and the trigger. At inference we execute 25 steps of each 50-step chunk, cap joint speed at 3° per step, and fire once the arm is still and the predicted trigger exceeds 0.5 for three steps.

## Data

| | |
|---|---|
| Demonstrations | 116 (29,131 frames, 16.2 min) |
| Cells | 23 of 25 (A5, E5 absent) |
| Trigger label | 1 for 10 steps after the button, else 0 |
| Variation | 23 live / 93 dry shots, 3 clutter levels, 17 start poses |
| Format | LeRobot v3.0, CC BY 4.0 |

## Training

We ran 13 fine-tunes on Physical Intelligence's hosted service, varying schedule, batch size, data mixture and base checkpoint (Table 1).

Table 1. Validation error by run. Std.: standard base; Alt.: alternative base offered by the service.

| Run | Base | Passes | Val. error |
|---|---|---|---|
| Baseline | Std. | 26.5 | 0.298 |
| **Longer schedule (demo)** | Std. | 33.5 | **0.287** |
| Much longer | Std. | 48.5 | 0.303 |
| Batch 512 | Std. | 48.5 | 0.334 |
| Logo cells oversampled | Std. | 33.5 | 0.333 |
| Final-crosshair takes only | Std. | ∼47 | 0.325† |
| Still frames dropped | Std. | 26.5 | 0.421‡ |
| Baseline | Alt. | 26.5 | 0.294 |
| Shorter schedule | Alt. | 17.6 | 0.294 |
| Logo cells oversampled | Alt. | 17.6 | 0.281§ |
| Final-crosshair takes only | Alt. | ∼47 | 0.294† |
| Still frames dropped | Alt. | 17.6 | 0.375‡ |
| Logo + still dropped | Alt. | 17.6 | 0.444‡ |

Best checkpoint per run. † Validated on a 101-take subset. ‡ Validated on moving frames only. § Validation split likely over-represents the logo cells.

## Findings

- **Schedule matters most.** Error is lowest at about 33 passes on the standard base and 18 on the alternative base; longer training overfits.
- **More data beat cleaner data.** Dropping the 15 takes recorded before the final crosshair did not help.
- **Batch size and oversampling did not help** on the standard base.
- **The policy follows the instruction.** With images and joint state held fixed and only the prompt's cell changed, all 24 probed checkpoints aimed in column and row order.
- **The trigger is the weak point.** Each take continues 2 s after the shot with the trigger at 0, so the policy rarely emitted a clean pulse offline; the stillness rule above compensates.

## Limitations

We did not measure a real-world hit rate; evidence is validation error, the prompt probe and a successful live demonstration. Camera and board positions were fixed, and coverage is uneven (two takes each for row 1, A3 and E3).

**Code and wiring:** `github.com/VineethSendilraj/pi07-dart-painting-so101`

[1] Physical Intelligence. $\pi_{0.7}$: A steerable generalist robotic foundation model with emergent capabilities. arXiv:2604.15483, 2026.

[2] L. X. Shi, B. Ichter, M. Equi, *et al.* Hi Robot: Open-ended instruction following with hierarchical vision-language-action models. ICML, 2025.

[3] R. Cadene, S. Alibert, A. Soare, *et al.* LeRobot: State-of-the-art machine learning for real-world robotics in PyTorch. `https://github.com/huggingface/lerobot`, 2024.